MIL-H-13514E
29 June 1983
SUPERSEDING
MIL-H-13514D
8 December 1972

MILITARY SPECIFICATION

HEATERS, SPACE, RADIANT TYPE, PORTABLE 45,000 BTU/HR.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers 45,000 BTU/HR, portable radiant type space heaters (see 6.1).

1.2 Classification. Heaters shall be of the following types, as specified (see 6.2):

Type I - Coal or solid fuel burning.
Type II - Oil burning.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-P-386	-	Pins, Cotter (Split)
QQ-S-781	-	Steel Strapping, Flat
QQ-W-461	-	Wire, Steel, Carbon, (Round, Bare and Coated)
PPP-C-843	-	Cushioning Material, Cellulosic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4520

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MILITARY

MIL-P-116	-	Preservation, Methods Of
MIL-P-551	-	Pipes and Elbows, Air Conditioning - Heating
MIL-P-584	-	Poker
MIL-D-1427	-	Damper, Flue, 4 Inch
MIL-B-2029	-	Burner, Assembly, w/Space Heater
MIL-C-10872	-	Cap, Flue, 4-Inch, Cylindrical
MIL-A-54003	-	Arrester, Spark, M-1941

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property

DRAWINGS

US ARMY NATICK RESEARCH AND DEVELOPMENT LABORATORIES

5-11-82	-	Heater, Space, Radiant Type, Portable, 45,000 BTU/Hr.; Stove Assembly and Details
5-11-83	-	Heater, Space, Radiant Type, Portable, 45,000 BTU/Hr.; Details "A"
5-11-84	-	Heater, Space, Radiant Type, Portable, 45,000 BTU/Hr.; Details "B"
5-11-85	-	Heater, Space, Radiant Type, Portable, 45,000 BTU/Hr.; Details "C"
5-13-1454	-	Heater, Space, Radiant Type, Portable, 45,000 BTU/Hr.; Container for Domestic and Level "A" Shipment

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A48	-	Gray Iron Castings
A109	-	Steel, Carbon, Cold-Rolled Strip
A366	-	Steel, Carbon, Cold-Rolled Sheet
D3951	-	Standard Practice for Commercial Packaging

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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of the document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2 and 6.3).

3.2 Materials and components. The materials and components shall be as specified herein and on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this document (see 6.4).

3.2.1 Steel.

3.2.1.1 Strip and sheet. Strip and sheet steel shall conform to ASTM A109 and A366 respectively.

3.2.1.2 Wire. Steel wire used for rivets shall conform to AISI 1010, finish 1, annealed of QQ-W-461.

3.2.2 Iron castings. Base metal for gray iron castings shall consist of not less than 40 percent pig iron and the remainder scrap iron. The castings shall have mechanical properties conforming to ASTM A48.

3.3 Design and construction. The design and construction of the type I and type II heaters shall conform to the drawings as follows, and as specified herein:

a. Type I - Applicable drawings, 5-11-82, 5-11-83, 5-11-84, and 5-11-85

b. Type II - Applicable drawings, 5-11-82, except grates are not required, 5-11-83 and 5-11-84

3.3.1 Type I heater. The type I heater shall consist of the components in table I. The casting of the grate assembly shall be clean and free of warpage, blow holes, cold shut and cracks. The surfaces shall have a uniform appearance with no visible evidence of projections and metal break out from the mold. The draw grate when assembled shall be capable of being pushed in and out with no interference.

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TABLE I. Components of type I heater

Quantity	Nomenclature
1	Base assembly
1	Grate assembly, round and draw
1	Adapter ring
1	Top assembly with lid assembly
6	Pipe, air conditioning - heating, straight, 4 inch diameter, 24 inches long, nested
1	Damper, flue
1	Poker, 24 inch
1	Shaker
1	Shovel, fire, hand
1	Arrester, spark

3.3.2 Type II heaters. The type II heater shall consist of the components in table II.

TABLE II. Components of type II heater

Quantity	Nomenclature
1	Base assembly
1	Adapter ring
1	Top assembly with lid assembly
6	Pipe, air conditioning - heating, straight, 4 inch diameter, 24 inches long, nested
1	Burner, assembly, space heater
1	Cap, flue, 4-inch, cylindrical

3.4 Pipe, air conditioning - heating, straight. The 4 inch diameter, 24 inch long, straight, air conditioning-heating pipe shall be a blued finish and conform to MIL-P-551. Six lengths of pipe shall be furnished with each heater.

3.5 Damper, flue. The flue damper shall conform to MIL-D-1427.

3.6 Poker, 24 inch. The 24 inch poker shall conform to MIL-P-584.

3.7 Pin, cotter. The cotter pin shown on Drawing 5-11-85 shall conform to type A of FF-P-386.

3.8 Burner, assembly. The oil burner assembly shall conform to MIL-B-2029.

3.9 Shovel, fire, hand. The shovel, fire, hand shall be furnished as Government furnished property (see 6.2 and 6.2.1).

3.10 Arrester, spark. The spark arrester shall conform to MIL-A-54003.

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3.11 Cap, flue, 4-inch, cylindrical. The 4-inch cylindrical cap flue shall conform to MIL-C-10872.

3.12 Marking for identification. Heaters shall be marked for identification in accordance with the unit marking requirements of MIL-STD-130 and as specified on applicable drawings.

3.13 Workmanship. Space heaters shall be clean and free of distortion and misalignment of parts. Spot welds shall be adherent and not burned through. All parts formed from sheet or strip steel shall be free of warpage, dents, wrinkles, bends not specified, fractures, sharp edges, burrs and slivers.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 End item visual examination. The end item shall be examined for the defects listed in table III. The lot size shall be expressed in units of one heater. The sample unit shall be one heater. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

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TABLE III. End item visual defects

Examine	Defect	Classification Major	Minor
Construction and workmanship	Accessory or component missing, separated, or malformed	X	
	Sheet metal fractured, dented, torn, wrinkled, or with open laminations	X	
	Sharp edges, burrs, or slivers		X
	Parts misaligned	X	
	Castings with prominent parting lines due to improper molding	X	
	Cracks	X	
	Blowholes, hard spots, cold shuts, or porosity	X	
	Not clean		X
	Projections or metal breakout from mold	X	
	Draw grate movement restricted due to interference	X	
Top assembly	Collar improperly formed or trimmed, top edge of uneven height, sharp or turned outward, or loose	X	
	Spot welds broken, missing, or cratered due to metal burning	X	
	Heater top does not seat into adapter ring	X	
	Lid lug bent, affecting use		X
	Lid distorted or opened around flanged edge		X
	Lid latch not free moving or turning		X
	Spot welds on handle less than two	X	
Adapter ring	Adapter ring does not seat properly in heater base, or distorted	X	
	Top edge of flange uneven		X
Base assembly	Door does not fit contour of front of base (opening between door and base)	X	
	Door latch or door loose		X
	Hinge not in proper location		X
	Door latch embossing missing or too small		X

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Base assembly (cont'd)	Door latch not free turning		X
	Door latch rivet improperly peened or not peened		X
	Door out of alignment		X
	Hinge binds		X
Grates	Warped more than 1/8 inch in any direction	X	
Marking for identification	Missing, illegible, imcomplete, wrong size, or wrong location	X	

4.4.3 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified on the drawings. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of one heater. The sample unit shall be one heater. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking comply with the section 5 requirements. Defects shall be as classified in accordance with table IV. The sample unit shall be one shipping container fully packaged. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

TABLE IV. Packaging defects

Examine	Defect
Markings	Omitted; incorrect; illegible; or of improper size, location, sequence, or method of application
Preservation	Preservation improperly applied or missing
Materials	Component missing or damaged
Workmanship	Inadequate application of components such as incomplete closure of container flaps; strapping, sealing, supports, braces, blocking or cushioning inadequate, improper, or missing; bolts and closures improperly affixed; components improperly secured or not arranged as specified; or inadequate strapping
	Bulged or distorted container

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Preservative application.

5.1.1.1.1 Type I and type II heaters. All uncoated ferrous metal surfaces of each heater and components, except the pipe sections shall be cleaned by method C-1, thoroughly dried and coated with type P-1 preservative in accordance with MIL-P-116. The pipe sections shall be coated with preservative in accordance with MIL-P-551.

5.1.1.2 Unit packing.

5.1.1.2.1 Pipe sections. Thirty sections of pipe, prepared as specified in 5.1.1.1.1, shall be unit packed in accordance with MIL-P-551, except that prior to wrapping the pipe sections for the type I heater, five pokers, five shakers, and five dampers, prepared as specified in 5.1.1.1.1, shall be placed in the center of the nested pipe sections and heater top (see 5.2.1.1.2). The size of the fiberboard box shall be such as to fit within the lid opening of the heater.

5.1.2 Commercial. The pipe sections, poker, shakers, and dampers shall be preserved in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A or Commercial as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Type I heater.

5.2.1.1.1 Heater bases. A cushioning pad approximately 12 inches by 12 inches by 1/4 of an inch, conforming to type II, class B of PPP-C-843 shall be placed in the bottom of each heater base. An assembled grate shall then be placed on top of the cushioning pad in each of the five heater bases. A U-support, constructed in accordance with Drawing 5-13-1454, shall then be placed over each grate so there will be no direct pressure on the grate when the remaining bases are nested therein. The five bases shall then be nested together. One base block shall then be placed in the bottom of the top nested base as specified in Drawing 5-13-1454, to insure a level foundation for the nested tops.

5.2.1.1.2 Heater tops. Three heater tops shall be inverted and two blocks, detailed on Drawing 5-13-1454, shall then be placed in each of the three inverted heater tops, one on each side of the lid opening. A lid shall then be secured in place in one of the tops, and the remaining two tops shall be nested in the top with the lid in place. A fourth top shall then be nested in place. The tops shall be

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nested with the lid openings in alignment. The four nested tops shall then be placed inside the fifth heater base. Five adapter rings shall then be nested together and placed in the nested tops. The fiberboard box containing the pipe sections unit packed as specified in 5.1.1.2, shall be placed on end, in the lid openings of the nested tops. Five shovels shall be positioned handles down, around the box. The remaining three lids shall be placed on edge beside the shovels. A fifth heater top, with lid secured in place, shall then be placed right side up on top of the nested rings. Extreme care shall be exercised during the nesting and packing operations to insure a firm and straight pack.

5.2.1.1.3 Complete heaters. The five complete type I heaters, prepared and arranged as specified above, shall be packed in a nailed wood crate constructed in accordance with Drawing 5-13-1454. One crossmember of the base of the crate shall be centered at the door of the bottom heater base. Each crate shall be tightly strapped with two 3/4 by 0.023-inch flat steel straps positioned at right angles to each other and around the top, sides, and base of the crate, and two 3/4 by 0.023-inch flat steel straps applied girthwise and located approximately 12 inches from each end of the crate. The flat steel strap shall conform to type I, class B of QQ-S-781. The straps shall be stapled as shown on Drawing 5-13-1454.

5.2.1.2 Type II heaters.

5.2.1.2.1 Heater bases. Five heater bases shall be nested within one another, separated by spacers constructed of two pieces of lumber, nominal 1 by 3 inch, 12 inches long, on edge, the end of one butted against the center of the face of the other and fastened with two nails to form a "T". Two top blocks, detailed on Drawing 5-13-1454, shall be placed, 1-5/8 inches face down, in opposite sides of the top nested base to insure a level foundation for the nested tops.

5.2.1.2.2 Pipe sections, heater tops, and adapter rings. The thirty pipe sections only, unit packed as specified in 5.1.1.2, the five heater tops, and five adapter rings shall be packed as specified in 5.2.1.1.2.

5.2.1.2.3 Complete heaters. The five complete type II heaters prepared and arranged as specified above, shall be packed as specified in 5.2.1.1.3.

5.2.2 Commercial packing. Type I and type II complete heaters preserved as specified in 5.1, shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any special marking required by the contract, interior packages and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D3951, as applicable. In addition to the marking as specified above, each crate containing five type II heaters shall be provided with two tags containing the following precautionary marking:

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"CAUTION

THIS ITEM CANNOT BE UTILIZED

WITHOUT

BURNER ASSEMBLY, SPACE HEATER"

Each tag shall be secured in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The space heaters are intended to heat tents and temporary shelters.

6.1.1 Type I. The type I space heater is intended for use with coal or other solid fuel and may be converted to an oil burning space heater by removing the grate assembly and inserting the oil burner covered by MIL-B-2029.

6.1.2 Type II. The type II space heater is furnished with the oil burner covered by MIL-B-2029.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this document.
(b) Type space heater required (see 1.2).
(c) When a first article is required (see 3.1).
(d) Source and delivery of Government furnished property (see 3.9 and 6.2.1).
(e) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

6.2.1 Government furnished property. Government furnished property should be obtained under NSN 5120-00-293-0450.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of one heater. The contracting officer shall include specific instructions in all acquisition documents regarding arrangements for inspection of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - YD
Air Force - 99

Preparing activity:

Army - GL

Project No. 4520-0175

Review activities:

Army - ME
Air Force - 84

User activities:

Army - CE
Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-13514E	Heaters, Space, Radiant Type, Portable 45,000 BTU/HR.

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 APR 83

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